Lie Theory: Chapter 6

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1. Let L be a complex Lie algebra. Show that L is nilpotent if and only if every 2-dimensional subalgebra of L is abelian. (Use the second version of Engel's Theorem.)

Proof. By Engel's Theorem, we can alternatively show that every 2-dimensional subalgebra is abelian if and only if for all $x \in L$, $\operatorname{ad} x$ is nilpotent.

If $\operatorname{ad} x$ is not nilpotent for some $x \in L$, then it has some nonzero eigenvalue λ. Let y be an eigenvector corresponding to λ. Then $\operatorname{span}\{x, y\}$ is a subalgebra and $[x, y] \neq 0$ by construction, so this is a two-dimensional subalgebra which is not abelian.

On the other hand, if there is a nonabelian 2-dimensional subalgebra $A \subseteq L$, by the classification of nonabelian 2-dimensional Lie algebras, we have a basis $\{x, y\}$ such that $[x, y] = y$, so $\operatorname{ad} x$ has 1 as an eigenvalue, meaning it cannot be nilpotent. □

2.

(i) Let L be a solvable Lie subalgebra of $\mathfrak{gl}(V)$, where V is a complex vector space. Show that every element of L' is a nilpotent endomorphism of V.

(ii) Let L be a complex Lie algebra. Show that L is solvable if and only if L' is nilpotent.

Proof.

(i) By Lie's theorem, there is a basis $x_1, \ldots, x_n$ where $n = \dim V$ under which every element of L is represented by an upper triangular matrix. As we have seen, this means that an element of L', being a linear combination of brackets of upper triangular matrices (and thus upper triangular itself since upper triangular matrices form a subspace) is strictly upper triangular, and thus nilpotent as a map, as desired.

(ii) If L' is nilpotent, then it is also solvable, so since $L^{(k+1)} = (L')^{(k)}$ and there is some j for which $(L')^{(j)} = 0$, we have $L^{(j+1)} = 0$ so L is solvable.

Now if L is solvable, then $\operatorname{ad} L$ is a solvable Lie subalgebra of $\mathfrak{gl}(L)$ since ad is a Lie algebra homomorphism. Thus every element of $(\operatorname{ad} L)'$ is a nilpotent endomorphism of L by (i). Now since $\operatorname{ad} L' = (\operatorname{ad} L)'$, by Engel's theorem we have L' is nilpotent.

□

3. Use Lie's Theorem to give another proof of Proposition 5.7.

4. Prove that $\frac{d}{dt} \exp(tA) = A \exp(tA)$.

Proof.

$$\begin{aligned}
\frac{d}{dt}\exp(tA) &= \frac{d}{dt}\left(1+\sum_{i=1}^{\infty}\frac{t^iA^i}{i!}\right)\\
&= \frac{d}{dt}1+\sum_{i=1}^{\infty}\frac{d}{dt}\frac{t^iA^i}{i!}\\
&= \sum_{i=1}^{\infty}\frac{it^{i-1}A^i}{i!}\\
&= \sum_{i=0}^{\infty}\frac{(i+1)t^iA^{i+1}}{(i+1)!}\\
&= \sum_{i=0}^{\infty}\frac{t^iA^{i+1}}{i!}\\
&= A\sum_{i=0}^{\infty}\frac{t^iA^i}{i!}\\
&= A\exp(tA)
\end{aligned}$$

□

5. Let A be an arbitrary square matrix. Prove that $\exp(A)$ is invertible and find its inverse.

Proof. We will show that the inverse of $\exp(A)$ is $\exp(-A)$. Define $f(t) = \exp(tA)\exp(-tA)$. We will show that when $t = 1$, $f(t)$ is the identity map. Then using the power rule together with problem 4, we have

$$\begin{aligned}
\frac{d}{dt}f(t) &= \frac{d}{dt}\exp(tA)\exp(-tA)\\
&= \left(\frac{d}{dt}\exp(tA)\right)\exp(-tA) + \left(\exp(tA)\frac{d}{dt}\exp(t(-A))\right)\\
&= A\exp(tA)\exp(-tA) + \exp(tA)(-A)\exp(-tA)\\
&= A\exp(tA)\exp(-tA) + (-A)\exp(tA)\exp(-tA)\\
&= A\exp(tA)\exp(-tA) - A\exp(tA)\exp(-tA)\\
&= 0.
\end{aligned}$$

where the fourth equality comes from realizing that $-A$ commutes with $\exp(tA)$ because it commutes with each term in the series defining $\exp(tA)$. Thus $f(t)$ is constant over t, so in particular its value at $t = 0$ is also its value at $t = 1$. Of course,

$$f(0) = \exp(0A)\exp(-0A) = \exp(0)\exp(0) = 1 \cdot 1 = 1,$$

so then we also have $f(1) = 1$ (the identity map), as desired. □

6. Let A be an arbitrary element of $\mathfrak{sl}(2, \mathbb{C})$. What can you say about the determinant of $\exp(A)$?

Proof. First we observe that for $A, P \in \mathfrak{sl}(2, \mathbb{C})$,

$$\begin{aligned} P \exp(A) P^{-1} &= P \left(\sum_{n=0}^{\infty} \frac{A^n}{n!} \right) P^{-1} \\ &= \sum_{n=0}^{\infty} \frac{P A^n P^{-1}}{n!} \\ &= \sum_{n=0}^{\infty} \frac{(PAP^{-1})^n}{n!} \\ &= \exp(PAP^{-1}), \end{aligned}$$

so conjugating before taking exp gives the same result as conjugating afterwards. Thus since

$$\begin{aligned} \det(P \exp(A) P^{-1}) &= \det(P) \det(\exp(A)) \det(P^{-1}) \\ &= \det(P) \det(P^{-1}) \det(\exp(A)) \\ &= \det(PP^{-1}) \det(\exp(A)) \\ &= \det(1) \det(\exp(A)) \\ &= 1 \cdot \det(\exp(A)) \\ &= \det(\exp(A)), \end{aligned}$$

$\exp(A)$ has the same determinant as any matrix similar to it (in the sense of being related by conjugation). Since every square matrix over $\mathbb{C}$ is similar to an upper triangular matrix, we henceforth treat A as upper triangular. Along its diagonal, we have the eigenvalues λ_1 and λ_2. Along the diagonal of A^n, we have λ_1^n and λ_2^n. Thus along the diagonal of $\exp(A)$, we have

$$\sum_{n=0}^{\infty} \frac{\lambda_i^n}{n!} = e^{\lambda_i}$$

for $i = 1, 2$. Thus

$$\begin{aligned} \det(\exp(A)) &= e^{\lambda_1} e^{\lambda_2} \\ &= e^{\lambda_1 + \lambda_2} \\ &= e^{\operatorname{tr}(A)}. \end{aligned}$$

Note that the above argument works for any square matrix; we have not used anything special about the dimension besides the number of eigenvalues, and we have not used anything special about the trace of A being 0. Now, however, we use $\operatorname{tr}(A) = 0$ to conclude that $\det(\exp(A)) = e^0 = 1$. □

7. Consider the orthogonal Lie algebra $\mathfrak{so}(3, \mathbb{C})$.

(a) Find a basis for $\mathfrak{so}(3, \mathbb{C})$.

(b) Compute $\exp(A)$ for each element A of this basis.

Proof.

(a) Recall that $\mathfrak{so}(3,\mathbb{C}) = \{x \in \mathfrak{gl}(3,\mathbb{C}) : x^t S = -Sx\}$ where

$$S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

If $x \in \mathfrak{gl}(3,\mathbb{C})$, note that $S^2 = I_3$ and $S^t = S$, so

$$(Sx)^t = x^t S^t = x^t S,$$

so $-Sx = (Sx)^t$, or in other words, Sx is skew-symmetric. In fact, noting that $S(Sx) = x$, if you take any skew-symmetric matrix and multiply it on the left by S, it satisfies the property $x^t S = -Sx$, so $\mathfrak{so}(3,\mathbb{C})$ **is** the set of 3×3 skew-symmetric matrices multiplied on the left by S. A basis for 3×3 skew-symmetric matrices is

$$\{e_{12} - e_{21}, e_{13} - e_{31}, e_{23} - e_{32}\}.$$

Since S just swaps the second and third rows, it is a bijection, and thus $y \mapsto Sy$ is a (linear) bijection, and we can map the above basis to a basis of $\mathfrak{so}(3,\mathbb{C})$ by multiplying on the left by S to get

$$\{e_{12} - e_{31}, e_{13} - e_{21}, e_{33} - e_{22}\}.$$

(b) We have

$$\begin{aligned} (e_{12} - e_{31})^2 &= -e_{32} \\ (e_{12} - e_{31})^3 &= -e_{32}(e_{12} - e_{31}) = 0, \end{aligned}$$

so $\exp(e_{12} - e_{31}) = 1 + e_{12} - e_{31} - e_{32}/2$,

$$\begin{aligned} (e_{13} - e_{21})^2 &= -e_{23} \\ (e_{13} - e_{21})^3 &= -e_{23}(e_{13} - e_{21}) = 0, \end{aligned}$$

so $\exp(e_{13} - e_{21}) = 1 + e_{13} - e_{21} - e_{23}/2$, and

$$\begin{aligned} (e_{33} - e_{22})^2 &= e_{33} + e_{22} \\ (e_{33} - e_{22})^3 &= (e_{33} + e_{22})(e_{33} - e_{22}) = e_{33} - e_{22}, \end{aligned}$$

so

$$\begin{aligned} \exp(e_{33} - e_{22}) &= 1 + (e_{33} - e_{22}) + (e_{33} + e_{22})/2 + (e_{33} - e_{22})/6 + \cdots \\ &= 1 - e_{33} - e_{22} + e_{33} \sum_{n=0}^{\infty} \frac{1}{n!} + e_{22} \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \\ &= 1 - e_{33} - e_{22} + e_{33} \cdot e + e_{22} \cdot e^{-1} \\ &= 1 + e_{33}(e-1) + e_{22}(e^{-1} - 1) \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & e^{-1} - 1 & 0 \\ 0 & 0 & e - 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & e^{-1} & 0 \\ 0 & 0 & e \end{pmatrix} \\ &= e_{11} + e^{-1} \cdot e_{22} + e \cdot e_{33}. \end{aligned}$$

□

8. Let p be prime and let F be a field of characteristic p. Consider the following $p \times p$ matrices:

$$x = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \end{pmatrix}, \qquad y = \begin{pmatrix} 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & p-2 & 0 \\ 0 & 0 & \dots & 0 & p-1 \end{pmatrix}.$$

Check that $[x, y] = x$. Deduce that x and y span a 2-dimensional solvable subalgebra L of $\mathfrak{gl}(p, F)$. Show that x and y have no common eigenvector, and so the conclusions of Proposition 6.6 and Lie's Theorem fail. Show that the conclusion of part (i) of #2 also fails.

Proof. Indeed,

$$\begin{aligned}
&[x, y] \\
&= \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & p-2 & 0 \\ 0 & 0 & \dots & 0 & p-1 \end{pmatrix} - \begin{pmatrix} 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & p-2 & 0 \\ 0 & 0 & \dots & 0 & p-1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & p-2 & 0 \\ 0 & 0 & \dots & 0 & p-1 \\ 0 & 0 & \dots & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & p-2 \\ p-1 & 0 & 0 & \dots & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1-p & 0 & 0 & \dots & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \end{pmatrix} \quad \text{(since } F \text{ has characteristic } p\text{)} \\
&= x.
\end{aligned}$$

Thus x and y span a 2-dimensional nonabelian solvable subalgebra L of $\mathfrak{gl}(p, F)$ by the classification of low-dimensional nonabelian Lie algebras.

Now for eigenvectors, note that since y is diagonal and each eigenvalue is distinct since F has characteristic p, the eigenvectors are multiples of the standard basis vectors e_i. Thus choose one such e_i corresponding to λ_i, and note that $ye_i = \lambda_i e_i$. Then $x(\lambda_i e_i)$ returns the

i^{th} column of x multiplied by λ_i. This cannot be a multiple of $\lambda_i e_i$, however, since x has the effect of moving the top row to the bottom, so $\lambda_i e_i$ is not an eigenvector of x, and thus x and y have no shared eigenvectors.

Finally, since $x = [x, y]$, $x \in L'$, but since it just shifts the top row to the bottom, it is not nilpotent, contradicting the conclusion of 2(i). $\square$